

Leo Kinyera

Biomedical Engineer and Data Science student with specialized experience in low-field MRI innovation, including hands-on development of open-source software tools (mri4all, MaRGE, Deep learning models, Pypulseq) and permanent magnet systems. Passionate about leveraging advanced MRI technologies for healthcare accessibility in resource-limited settings but also so interested in developing AI systems of healthcare.

Research Interests

Biomedical Engineering • Low-field MRI innovation • Deep learning for medical imaging • Open-source healthcare technology • PCB and circuit design • Democratizing AI in resource-limited settings • MRI Physics

Educational Background

BS in Biomedical Engineering

Mbarara University of Science and Technology
P.O. Box 1410, Mbarara Uganda
2021-2025

Ordinary Level

St. JohnBosco Seminary Hoima
P.O. BOX 139 Hoima, Uganda
2015 - 2018

Biology Chemistry Mathematics (Advanced Level)

Mandela Secondary School
P.O. BOX 409 Hoima, Uganda
2019 - 2020

Primary School

St. Francisco Primary School
P.O. BOX 224 Hoima, Uganda
2004 - 2014

Research & Academic Experience

Building an image enhancement model for low-field MRI

August 2024 - December 2025

- I brought together a team of deep learning experts from Havard and MIT, including top radiologists from Makerere University for this project.
- Collaborated with the i3M-MRI lab from Valencia Spain for dedicated Tesla A100 NVIDIA GPUs to aid our team build our model without any compute power disruptions.

Volunteering at MRI Uganda

May 2022 - August 2025 (~3 years)

- Led the transition from command-line Python code to the user-friendly mri4all GUI, significantly improving usability and diagnostic efficiency for MRI prototypes.
- Co-authored a conference paper - "Democratizing AI in Africa: Federated Learning for Low-Resource Edge Devices"
- I am currently actively developing a deep learning model to enhance image quality for our MRI system.

Publications & Conference Contributions

- "Democratizing AI in Africa: Federated Learning for Low-Resource Edge Devices" – Conference paper (Co-author).
- First AI for Health Symposium, Mbarara University of Science and Technology (2024) – Participant.
- Second AI for Health Symposium, Mbarara University of Science and Technology (2025) – Participant and Speaker.

Contact

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Skills

Technical Skills

- Software development
- CAD
- Electronics
- Data Science and Analytics
- Research collaboration and technical documentation in low-resource MRI settings.
- PCB design

Tools/Software

- Github
- MaRGE
- MaRCoS
- PyPulseq
- SOLIDWORKS
- COMSOL/ANSYS
- Python
- Altium

Soft skills

- Effective communication skills
- Project management
- Team work
- Public Speech
- Effective presentation skills

Languages

- English (Proficient)
- Acholi (Fluent)
- Runyoro (Proficient)

Certification/Short Courses

- Software Engineering Certificate, ALX-Africa
- JavaScript programing, Coursera
- Docker for containerization, datacamp
- Kafka for event driven systems, datacamp

Note:

- My other co-author publications can be found on my google scholar profile in my full name.

Teaching & Mentorship

MRI Uganda Lab Internship June to August 2025

- Supervised the Software interns in laboratory projects and activities
- Trained both the software and Hardware teams on MRI hardware and software
- Create twelve software coding labs using Google Colab that shall be used in yearly internships for those to come.

Biomedical Engineering Faculty, MUST (April 2025)

- Co-tutored the Biomedical Engineering Workshop (Year 2–4 students).
- Delivered sessions on circuit design, PCB design, biomedical signal acquisition (PPG).
- Assisted in MRI practical sessions for final-year students (Imaging II).

Projects

Enhanced UI/UX of MaRGE

June 2025 - July 2025

- During the internship period at MRI Uganda (June to August 2025), I added a modern look and feel to the MRI console GUI MaRGE. I did that by adding over 15 themes and a number of font styles. I also improved its navigation so that the user no longer unnecessarily needs to manage many windows simultaneously.

Semi-autotuning and matching circuit

November 2024 - May 2025

- Created a team of three and designed a semi-autotuning and matching circuit for our low-field MRI prototype to reduce the imaging protocol time for phantoms and any imaging subjects.

Built a 9-ring 47mT permanent magnet system

August 2024 - September 2024

- Designed and simulated a 9-ring 47mT permanent magnet system using SOLIDWORKS, COMSOL, and ANSYS. Successfully integrated MaRGE for operational functionality, contributing to advancements in low-field MRI system development.

Installation of mri4all

2023

- Installed and configured mri4all for MRI-Uganda's prototype, enabling efficient pulse sequence execution and improving system calibration processes.

Professional Memberships

Rotary Uganda - Mbarara Branch

- Joined in 2022 during my first year at Mbarara University
- Did community service

Mbarara University Biomedical Engineering Students' Association (MBUBESA)

- Served as Academic Co-ordinator 2023 - 2024
- Launched an open-source Deep learning project for Biomedical Engineering students, titled ML-Algo-Exploration

International Society of Magnet Resonance Medicine (ISMRM)

- Joined October 2024
- Joined the Body MRI study group
- Joined the Brain Function study group
- Joined the Artificial Intelligence / Machine Learning study group

Referee

Referee 1:

Dr. Johnes Obungoloch

Faculty Dean, Faculty of Applied Science and Technology.

Mbarara University of Science and Technology

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Referee 2:

Leo Anthony Celi, MD, MS, MPH

Clinical Research Director and Senior Research Scientist, Laboratory for Computational Physiology, MIT

Associate Program Director, Internal Medicine Residency, Beth Israel Deaconess Medical Center

Associate Professor, Harvard Medical School (Part Time)

Editor-in-Chief, PLOS Digital Health

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